Same Deadline, Different Persistence

Submission timing and post-deadline activity among students with comparable clicks

Group 9: Jiayang Liu (50027216), Yanbin Chen (50026846),
Yanshun Wang (50026688), Yuming Zhang (50025933)

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Page-count note. The main text ends on page 7; references, disclosures and the appendix are excluded from the 6–10 page main-text requirement.

1 Question

Online course platforms record every resource view with a timestamp. Total click volume is the obvious summary of that record and the obvious baseline for predicting who will fail, yet two students can click the same number of times and finish very differently. This project asks whether the timing of activity around a mandatory deadline carries information that the volume alone does not. The specific question is: within the same module presentation, among students whose clicks in the 14 days after the first tutor-marked assignment (TMA) deadline are comparable, are submission timing and post-deadline persistence associated with eventually failing the module rather than passing or earning a distinction? Fail is the positive class, so the measure of interest is the average precision (AP) of Fail predictions against the Fail share of the same test records. Final Withdrawn records are a separate outcome: they leave the binary comparison and return only in the at-risk description of Section 6. Three choices define the design. First, the anchor is the deadline d of each presentation's earliest TMA with positive weight, not the start of the course, because the deadline is the moment at which a large share of students either submit or disappear. Second, the observation window is symmetric: 14 days before $(d - 13 \dots d)$ give a starting-point baseline, and 14 days after $(d + 1 \dots d + 14)$ give the behaviour of interest. Third, the comparison is paired on post-deadline clicks, so a difference in persistence cannot be explained by one group simply clicking more. Everything reported here is an observational association. The pairing is constructed from the final outcome, so the resulting sample is 1:1 by design and cannot estimate how many students fail. Deadlines are the scheduled dates in the assessment table; OULAD records neither individual extensions nor mark-release dates, so a submission is only observed as early or late relative to the scheduled date, and no score is treated as known at the cutoff.

2 Data and design

The analysis uses the Open University Learning Analytics Dataset (OULAD), the public release that accompanies the module-level records of 32,593 student enrolments in seven modules and 22 presentations. The unit of analysis is one enrolment (module × presentation × student), because the

Sample-flow step	Records
all_enrolments	32593
has_first_weighted_tma	30059
registered_by_deadline	29990
registered_at_cutoff	24594
banked_excluded	469
withdrawn_at_cutoff	4319
binary_records	19806
fail_records	6092

Table 1: Ordered eligibility flow. A record must carry a first weighted TMA, be registered by the deadline, still be enrolled at $d+14$ and not be a banked attempt. Withdrawn records drop out of the binary comparison but remain available to Section 6.

Module	Presentation	d	Enrolments	Eligible	Fail	Pass/Dist.
AAA	2013J	19	383	367	43	278
AAA	2014J	19	365	329	42	245
BBB	2013B	19	1767	1511	455	798
BBB	2013J	19	2237	1774	495	1056
BBB	2014B	12	1613	1242	376	716
BBB	2014J	54	2292	1733	391	1152
CCC	2014B	32	1936	1392	375	663
CCC	2014J	32	2498	1879	400	1000
DDD	2013B	25	1303	1106	352	506
DDD	2013J	25	1938	1572	403	815
DDD	2014B	25	1228	976	246	470
DDD	2014J	20	1803	1391	333	769
EEE	2013J	33	1052	908	200	609
EEE	2014B	33	694	595	164	356
EEE	2014J	33	1188	982	189	679
FFF	2013B	19	1614	1438	419	782
FFF	2013J	19	2283	1915	486	1081
FFF	2014B	24	1500	1243	366	638
FFF	2014J	24	2365	1772	357	1101

Table 2: The nineteen analysed presentations. Deadlines fall between day 12 and day 54 of the course, so a fixed course-day cutoff would compare students at very different points of the TMA cycle; anchoring on d removes that mismatch.

same person can appear in several presentations and the deadline is a property of the presentation. Of 32,593 enrolments, 24,125 satisfy every inclusion rule; 4,319 of those end in Withdrawn, and 19,806 have a binary outcome (6,092 Fail against 13,714 Pass/Distinction), so the Fail share is 30.8%. Nineteen presentations across modules AAA–FFF carry a positive-weight TMA; module GGG has none and never enters the design. Timing features come from the daily click stream. For every record the pipeline counts clicks per relative day between $d - 13$ and $d + 14$, then derives post-deadline active days, the longest run of consecutive zero-click days, the share of post-deadline clicks in the second week, the peak seven-day share, a normalised entropy of the daily distribution, and the day of first post-deadline activity. Submission fields are masked to $d + 14$: a submission recorded later is treated as not submitted, and two indicators plus the signed distance in days form the submission block.

Tolerance	Binary	Fail	Pairs	Fail coverage	Zero-click pairs	Mean click gap	Pre-click SMD
0%	19806	6092	4118	67.6%	607	0.0000	-0.306
5%	19806	6092	4768	78.3%	607	0.0020	-0.287
10%	19806	6092	4858	79.7%	607	0.0035	-0.279

Table 3: Pairing on post-deadline clicks. The 5% row is the primary specification; 0% and 10% are sensitivities. The pre-click standardised difference is reported because pairing deliberately does not balance the pre-deadline window.

Primary indicator	Pairs	Fail − Pass	95% interval	p	p_{Holm}	Reject
submitted_by_cutoff	4768	-0.1235	[-0.135, -0.112]	0.0010	0.0060	yes
submitted_after_due	4768	0.0839	[0.072, 0.096]	0.0010	0.0060	yes
relative_submission_day	3859	1.3809	[1.182, 1.568]	0.0010	0.0060	yes
post_active_days	4768	-0.3683	[-0.440, -0.295]	0.0010	0.0060	yes
post_longest_inactive_run	4768	0.3888	[0.303, 0.473]	0.0010	0.0060	yes
post_week2_share	4768	-0.0368	[-0.050, -0.024]	0.0010	0.0060	yes

Table 4: The six pre-specified paired differences, with a student-component bootstrap interval and Holm adjustment across the family. A negative value means Fail records score lower. The submission-distance indicator is only defined when both sides submitted by the cutoff, which is why its pair count is smaller. Intervals are conditional on the realised pairing.

3 Matched comparison

Pairing runs inside each presentation. Fail records are matched 1:1 with Pass/Distinction records whose post-deadline click total is first exactly equal and then within the tolerance; positive counts must satisfy a ratio bound, and zero-click records may only pair with other zero-click records. No record is reused. The greedy order is not claimed to be globally optimal, and the realised pairing is reported so the residual imbalance can be inspected. The primary match pairs 4,768 Fail records, covering 78.3% of all Fail records, with a mean relative click gap of 0.0020. Post-deadline volume is balanced almost exactly (standardised difference -0.000), but the pre-deadline window is not: its standardised difference is -0.287, and the raw difference is -20.6 clicks per record. Fail records were already somewhat less active before the deadline, so the paired difference answers a conditional question: given similar post-deadline clicks, what else differs? All six indicators point the same way and none of them is close to zero. Fail records are less likely to have submitted by the cutoff and more likely to submit late; among those who submitted they submit later; after the deadline they are active on fewer days, silent for longer stretches, and less active in the second week. Holm adjustment leaves every indicator significant, so the direction of the story does not depend on which of the six is treated as primary. The descriptive pattern is steep: 75.9% of records without a submission by $d + 14$ fail, against 15.9% of those who submitted a week or more early. This is descriptive, because submission and outcome both reflect the same underlying effort, but it motivates keeping submission status in every model. Requiring pre-deadline clicks within 20% keeps only 13.9% of the main pairs (664 pairs). The active-day difference stays negative (-0.398 days), so the post-deadline pattern is not simply an artefact of pre-deadline differences, but the subset is small and its interval is wider.

Submission band	Records	Fail	Pass/Dist.	Withdrawn	Fail share
not_submitted_by_14	2966	1507	478	981	75.9%
early_7plus	1667	240	1270	157	15.9%
early_1to6	11959	2056	8371	1532	19.7%
due_day	5373	1558	2789	1026	35.8%
late_1to7	1641	565	610	466	48.1%
late_8to14	519	166	196	157	45.9%

Table 5: Outcome mix by submission band in the eligible cohort. The band is built from the masked submission day, so a record without a submission by $d + 14$ counts as not submitted even if it submitted later.

Pre-click tolerance	Indicator	Pairs	Share of main	Fail − Pass
20%	submitted_by_cutoff	664	13.9%	-0.081
20%	submitted_after_due	664	13.9%	0.071
20%	relative_submission_day	580	13.9%	0.979
20%	post_active_days	664	13.9%	-0.398
20%	post_longest_inactive_run	664	13.9%	0.346
20%	post_week2_share	664	13.9%	-0.063
50%	submitted_by_cutoff	1331	27.9%	-0.056
50%	submitted_after_due	1331	27.9%	0.079
50%	relative_submission_day	1187	27.9%	1.048
50%	post_active_days	1331	27.9%	-0.322
50%	post_longest_inactive_run	1331	27.9%	0.284
50%	post_week2_share	1331	27.9%	-0.040

Table 6: Sensitivity that also requires the pre-deadline clicks to be close. The subset is selected after the fact and loses most of the sample, so it cannot replace the primary comparison; it shows whether the direction survives a stricter starting-point balance.

4 Predictive information

The pairing describes who differs at comparable clicks. A separate question is whether timing adds predictive information beyond volume and submission status. The ladder is fitted on a grouped 60/20/20 split (no student spans two parts), the threshold is chosen on the validation part, and every model is refitted on train plus validation before the test part is scored. The primary model reaches Fail AP 0.6611 against a no-skill floor of 0.3078. The ladder shows where the gain comes from: M0 already contains submission status and reaches 0.6544, the post-deadline timing block moves it to 0.6574, and the submission-distance terms add the rest up to 0.6611. Most of the predictive signal is submission status rather than the fine structure of the click sequence. A gradient-boosted alternative reaches 0.6649, within noise of the logistic ladder, so there is little evidence of a strong nonlinear pattern that the linear models miss. The threshold rule actually used is recorded per model (M0: target_precision, M1: target_precision, M2: target_precision, M3: target_precision, RF_M0: target_precision, RF_M2: target_precision). The pipeline first looks for a threshold that reaches 0.80 precision on the validation part and falls back to the maximum-F1 threshold when no such point exists, so the reported precision and recall describe a realisable operating point rather than the maximum of the precision-recall curve. With the label shuffled, the same pipeline returns Fail AP 0.3088 against a shuffled prevalence of 0.3069. A learning signal that survived shuffling would indicate leakage; it does not. Moving the cutoff from $d + 1$ to $d + 14$ raises the primary model's AP from 0.6449 to 0.6604 with the same feature list. The gain is therefore information gained by observing longer, not a better model.

Indicator	Pairs	Fail − Pass
submitted_by_cutoff	4161	-0.066
submitted_after_due	4161	0.096
relative_submission_day	3637	1.436
post_active_days	4161	-0.422
post_longest_inactive_run	4161	0.446
post_week2_share	4161	-0.042

Table 7: Sensitivity restricted to pairs where both records clicked at least once after the deadline, which removes the zero-click pairs.

Model	Features	Fail AP	No-skill AP	ROC-AUC	Brier	Cal. slope	F1
M0	7	0.6544	0.3078	0.7585	0.1658	0.909	0.472
M1	14	0.6574	0.3078	0.7606	0.1651	0.912	0.477
M2	16	0.6611	0.3078	0.7647	0.1641	0.918	0.475
M3	36	0.6608	0.3078	0.7653	0.1640	0.912	0.470
RF_M0	7	0.6541	0.3078	0.7576	0.1904	0.897	0.473
RF_M2	16	0.6649	0.3078	0.7681	0.1853	0.944	0.464

Table 8: Model ladder on the grouped main test split (five-seed mean). M0 uses pre- and post-deadline volume plus submission status; M1 adds post-deadline timing; M2 adds submission distance and is the primary model; M3 adds resource composition; M4 adds the first-TMA score and is a retrospective upper bound only. Thresholds are chosen on the validation part.

5 Validation design

The temporal split trains on the 2013 presentations and tests on 2014, dropping from the training side any student who also appears in 2014. Because most students take one module, this is a genuine forward check in time, but it is not an external dataset: the modules and the grading regime are the same. The leave-one-module-out folds address the other direction, generalising to a module the model has not seen, and the module composition of the test part differs a lot between folds.

6 Robustness and scope

All 5 of the indicators shown keep the same sign in every one of the eight specifications, including the two whole-window matching variants that balance a different quantity. The magnitude moves: matching on total clicks leaves less post-deadline difference, which is expected because the whole-window total is partly a pre-deadline quantity. The at-risk description is the only place where Withdrawn is modelled. Reading the binary comparison as a withdrawal warning would be wrong: those records were still enrolled at the cutoff by construction, and a model that predicts Fail among survivors is not the model that should flag a student who may leave.

7 Limitations

- AP is average precision for Fail, not accuracy. The no-skill reference is the Fail share in the same test records.

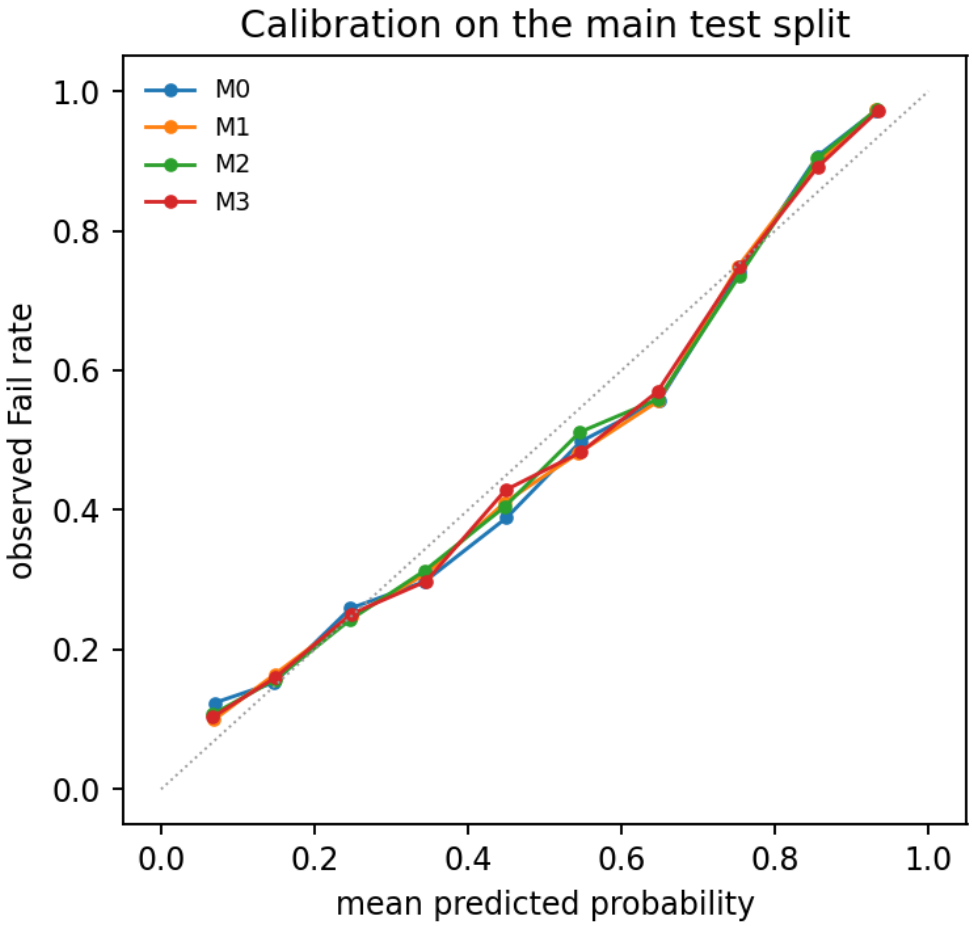


Figure 1: Reliability of the fitted probabilities on the main test split. Points near the diagonal are better calibrated; the ladder is not recalibrated after fitting, so miscalibration here is a property of the fitted model rather than a post-hoc correction.

- Pairing is 1:1 by design and cannot estimate the population failure rate. The greedy order is not globally optimal, and the bootstrap resamples students on the realised pairs only, so re-matching uncertainty is excluded.
- The binary cohort excludes records that withdrew before the cutoff. It is not a real-time early-warning population.
- The first-TMA score contributes to the final result. Models that use it are retrospective upper bounds, and OULAD records no mark-release date.
- The OULAD splits have been inspected across several rounds of exploration. The intervals and increments are exploratory even though the protocol is now frozen.
- VLE activity types are platform categories, not content quality; `studentVle` records day-level clicks only, so study time cannot be measured.

Two design facts deserve emphasis. First, the pre-deadline window is not balanced by the primary match; the sensitivity that balances it keeps only a small share of the pairs, so the honest statement

Post days	Model	Fail AP	Brier
1	M0	0.6380	0.1688
1	M1	0.6380	0.1688
1	M2	0.6449	0.1671
7	M0	0.6419	0.1685
7	M1	0.6414	0.1685
7	M2	0.6465	0.1672
14	M0	0.6536	0.1658
14	M1	0.6568	0.1652
14	M2	0.6604	0.1642

Table 9: Observation-window curve on the same split. A longer window supplies more information, so AP rises; the change is not a method gain and must not be read as one.

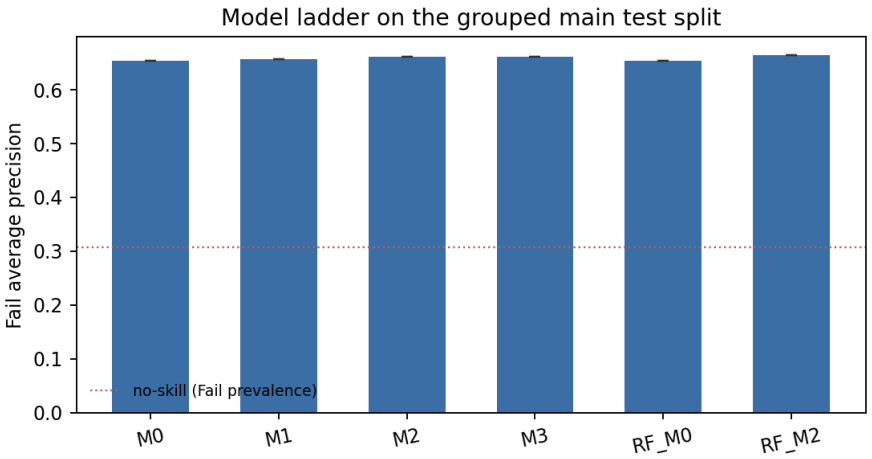


Figure 2: Model ladder on the grouped main test split; whiskers span the five seeds and the dotted line is the no-skill level.

is that the comparison holds post-deadline volume fixed and reports how much the pre-deadline history still differs. Second, the gain from timing features is small. Post-deadline persistence distinguishes outcomes descriptively far more clearly than it improves prediction, and the model that improves most adds submission distance, which is itself deadline behaviour rather than a click pattern.

Disclosures

AIGC tools assisted with code scaffolding, text editing and LaTeX formatting. All analysis code, numerical results and claims were reviewed by the group, and every figure and table is regenerated by the pipeline from the public dataset. No external collaborator contributed data or results.

Data and code availability

The unmodified OULAD archive is committed with its SHA-256. Extracted CSVs, student-level features, pair indices and predictions stay outside version control. One command reproduces the

Split	Train	Test	Train Fail	Test Fail	Student overlap
main_train_test	11892	3944	30.7%	30.8%	0
main_train_val	11892	3970	30.7%	30.9%	0
temporal	8162	11028	33.2%	29.4%	0
group_kfold_0	15844	3962	30.7%	31.2%	0
group_kfold_1	15845	3961	30.7%	31.1%	0
group_kfold_2	15845	3961	30.8%	30.4%	0
group_kfold_3	15845	3961	30.9%	30.3%	0
group_kfold_4	15845	3961	30.7%	30.8%	0
lomo_AAA	19196	608	31.3%	14.0%	0
lomo_BBB	14362	5439	30.5%	31.6%	0
lomo_CCC	16216	2438	31.5%	31.8%	0
lomo_DDD	15512	3894	30.1%	34.3%	0
lomo_EEE	17067	2197	31.1%	25.2%	0
lomo_FFF	14294	5230	30.6%	31.1%	0

Table 10: Every split is student-disjoint: no student appears on both sides. The overlap column is zero everywhere by construction and is checked at run time. The leave-one-module-out folds number six, not seven, because only six modules have a positive-weight TMA; each fold also drops the students who appear in the held-out module, so the training side shrinks.

analysis; the submission ZIP contains the code, the frozen configuration and the run log.

A Reproduction

```
make test
make sim && make validate RUN_ID=$(cat .last_sim_run)
make full && make validate RUN_ID=$(cat .last_full_run)
```

Every stage writes a stage summary, a handoff note and a JSONL event; the validator checks the log against the schemas in `docs/contracts` and reports zero errors and zero warnings.

Seed	Model	Test records	Fail share	Fail AP	Brier
42	M0	3942	0.3128	0.6659	0.1638
42	M1	3942	0.3128	0.6682	0.1632
42	M2	3942	0.3128	0.6715	0.1623
143	M0	3967	0.3166	0.6735	0.1639
143	M1	3967	0.3166	0.6755	0.1634
143	M2	3967	0.3166	0.6792	0.1625
144	M0	3958	0.3052	0.6539	0.1641
144	M1	3958	0.3052	0.6580	0.1629
144	M2	3958	0.3052	0.6572	0.1634
145	M0	3948	0.3057	0.6602	0.1627
145	M1	3948	0.3057	0.6629	0.1623
145	M2	3948	0.3057	0.6657	0.1616
146	M0	3951	0.3088	0.6778	0.1593
146	M1	3951	0.3088	0.6798	0.1586
146	M2	3951	0.3088	0.6835	0.1581

Table 11: Replication of the published two-week ladder: an 80/20 grouped holdout with the seed set of the exploratory study and no validation part. These values are the direct comparison point for the earlier tables; the main split above answers a slightly different question with a different partition.

Specification	Post days	Tolerance	Matched on	Pairs	Fail coverage	Pre SMD
primary	14	5%	post_clicks	4768	78.3%	-0.287
tolerance_0	14	0%	post_clicks	4118	67.6%	-0.306
tolerance_10	14	10%	post_clicks	4858	79.7%	-0.279
window_21_post	21	5%	post_clicks	4621	75.9%	-0.215
window_28_post	28	5%	post_clicks	4391	72.1%	-0.152
window_21_total	21	5%	total_clicks	4627	76.0%	0.051
window_28_total	28	5%	total_clicks	4477	73.5%	0.077
dedupe_events	14	5%	post_clicks	4779	78.4%	-0.268

Table 12: Robustness curve. The +21/+28 variants appear twice: matched on post-deadline clicks, which is comparable with the main design, and matched on whole-window clicks, which is comparable with the earlier +21/+28 study. The two answer different questions and must not be ranked against each other.

Indicator	Specifications negative	Specifications	Share negative
post_active_days	8	8	100.0%
post_longest_inactive_run	0	8	0.0%
post_week2_share	8	8	100.0%
submitted_after_due	0	8	0.0%
submitted_by_cutoff	8	8	100.0%

Table 13: Sign stability of each indicator across the eight specifications. A direction that flipped between reasonable choices would have to be reported as unstable; these do not flip.

Cluster	Records	Post clicks	Active days	Longest gap	Submitted by $d + 14$	Fail share
0.0	745	143.0	7.56	3.10	97.4%	47.0%
1.0	507	32.5	2.58	7.47	89.0%	55.6%
2.0	348	31.4	2.36	8.13	90.2%	48.3%

Table 14: Outcome-blind timing clusters inside the paired sample. k and the feature list were chosen without the final result. The largest cluster is also the most active, so cluster membership still tracks volume; the clusters are descriptive and are not a finding about learning styles.

Final outcome in the at-risk set	Records	Share
Fail	6092	25.3%
Pass	13714	56.8%
Withdrawn	4319	17.9%

Table 15: The at-risk set keeps Withdrawn as its own class. It is a different population from the binary comparison and answers a different question.